Transfer: waves and signals

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Choose appropriate wave models and explain what a signal establishes.

SUCCESS CRITERIA - I can...

1. Use wave speed and round-trip geometry correctly.
2. Separate frequency, amplitude and direction effects.
3. State the limits of an inference from a signal.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “frequency” and “wavelength” to explain your model or result. Sentence starters are optional.

frequency: number of cycles each second

wavelength: distance between corresponding points on repeated waves

round trip: travel from start to a reflector and back

01 Warm-up: bring back an earlier idea

A radar echo’s complete outward-and-return path is 5000 m.
Find the target range and explain the division.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $v = f\lambda$; $T = 1/f$; $d_{range} = v\,\Delta t/2$.
d_{range} is one-way range from round-trip time. Match speed and model to the wave and medium.

Worked reasoning: An echo time measures range using travel speed, while a frequency shift can indicate radial motion. Different observations answer different questions about the same target.

Your turn: Draw a measurement plan labeling outgoing pulse, return pulse and the time interval.

Sentence starter: My design shows … . It meets the requirement by … .



Marzano check - circle: ① ② ③

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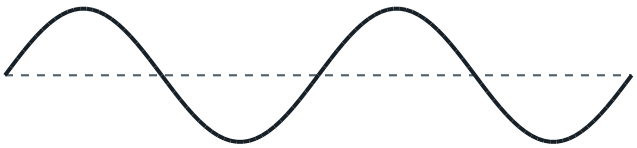
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Sketch a wave and label amplitude and wavelength 0.
Explain which tells about oscillation size and which about spatial repetition.

Sentence starter: My claim is … . The diagram/data show …, so … .



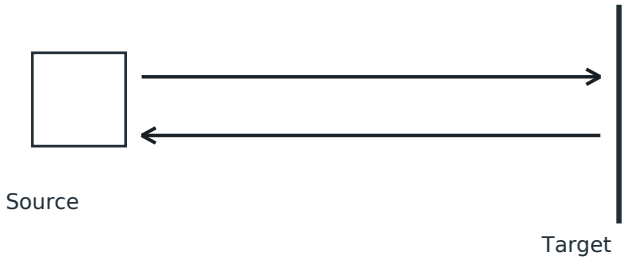
Sketch and label your response.

Marzano check - circle: 1 2 3

04 Apply the physics

A wave simulator uses frequency 5 Hz and wavelength 2 m.
Find propagation speed. An echo using that same speed returns after 4 s.
Label the round trip and calculate the target range.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: 1 2 3

05 Explain your reasoning

A source gets louder while its frequency remains fixed. Does pitch necessarily rise?
Explain using wave properties.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

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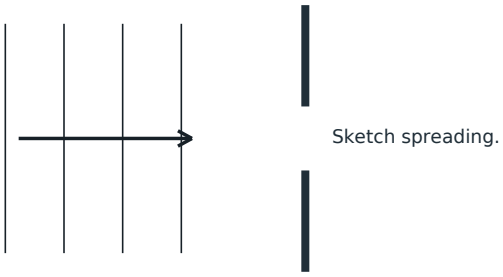
Continue down each column.

Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

Two waves meet the same opening with wavelengths 4 cm and 6 cm. Which generally diffracts more?
Explain the relevant comparison.
Label wavelength and opening width; sketch the emerging wavefronts.

Sentence starter: My claim is … . The diagram/data show …, so … .

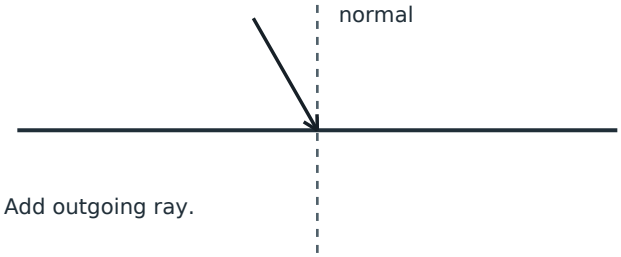


Marzano check - circle: 1 2 3

07 Exit check

A ray enters a slower medium at an angle.
Sketch whether it bends toward or away from the normal and name what happens to frequency.

Sentence starter: The … in my model represents … . Its label shows … .



Marzano check - circle: 1 2 3

08 STRETCH: test the model

A sound echo arrives after 0.5 s and a light echo after 2 microseconds.
Use 340 m/s and 3×10^{8} m/s. Which target is farther?

Sentence starter: I notice … . This supports … because … .



Marzano check - circle: 1 2 3